Forces & Equilibrium

ClipSAT Printable Practice Suite (clipsat.org) • Topic 12.01

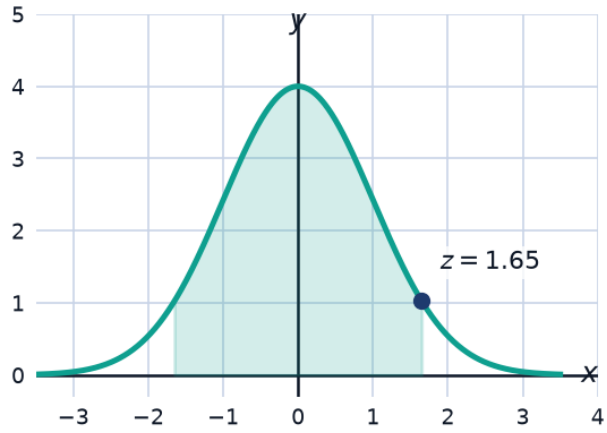
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Solve the inequality $x^2 + 17x + 72 > 0$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A block rests on a rough horizontal surface with normal reaction $N = 42$ N. The coefficient of friction is $\mu = 7/20$. Find the limiting frictional force.



3 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A small particle of weight 24 N is held in equilibrium by two light inextensible strings. String A makes an angle of 30° with the upward vertical, and string B makes an angle of 60° with the upward vertical. Find the tensions T_A and T_B in the two strings.

4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A block of mass 8 kg rests in limiting equilibrium on a rough plane inclined at 30° to the horizontal. Taking $g = 9.8$ m/s^2: (a) calculate the normal reaction R, (b) find the coefficient of friction μ between the block and the plane.

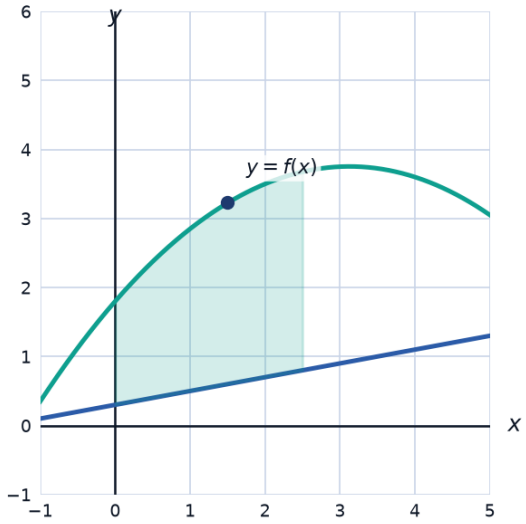
5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

[Forces & Equilibrium · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

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6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

In a circle of radius 10 cm, chord AB subtends an angle of $\theta = \frac{2\pi}{3}$ radians at the centre O. Find: (a) the exact length of chord AB, (b) the exact area of the minor segment bounded by chord AB and minor arc AB, in terms of π and $\sqrt{3}$.



7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The circle C has equation $x^2 + y^2 - 6x + 4y - 12 = 0$. (a) Find the coordinates of the centre and the exact radius of C. (b) Find the equation of the tangent to C at the point $P(6, 2)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the set of values of the constant k for which the equation $kx^2 + 4x + (k - 3) = 0$ has real roots.

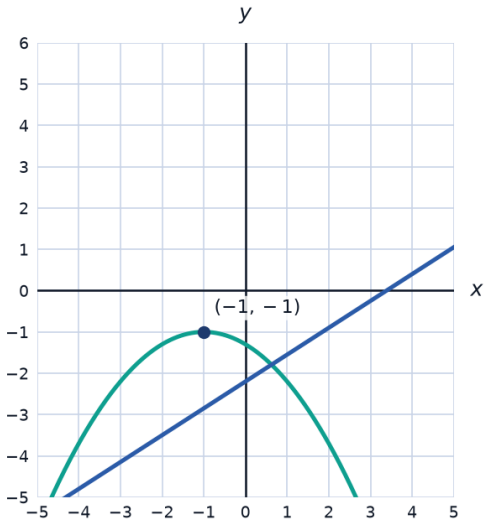
9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the equation of the normal to the curve $y = \frac{4}{\sqrt{x}} + 2x$ at the point where $x = 4$.

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10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A bag contains 5 red and 4 green marbles. Three marbles are drawn at random without replacement. Find the probability that: (a) all three are red, (b) at least two are red.



11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A block of mass $4\ \text{kg}$ is pulled along a rough horizontal plane ($\mu = 0.25$) by a pulling force of magnitude $25\ \text{N}$ acting at 30° above the horizontal. Taking $g = 9.8\ \text{m/s}^2$: (a) show that the normal reaction $R = 26.7\ \text{N}$, (b) find the acceleration of the block.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

The function f is defined by $f(x) = x^2 - 6x + 13$ for $x \geq 3$. (a) State the range of f. (b) Find an expression for $f^{-1}(x)$ and state the domain and range of f^{-1}.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Functions g and h are defined by $g(x) = 2x - 1$ for $x \in \mathbb{R}$, and $h(x) = \frac{6}{x-2}$ for $x \neq 2$. (a) Find an expression for $hg(x)$ in simplest form. (b) Solve the equation $hg(x) = g(x)$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the exact area of the region enclosed between the curve $y = 4x - x^2$ and the line $y = x$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

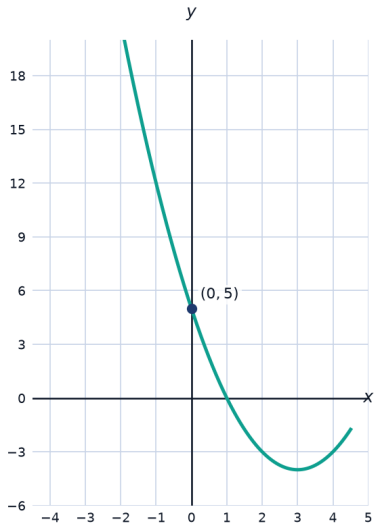
15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 9.01 Circular Measure

A curve has equation $y = 2x^3 - 9x^2 + 12x + 1$. (a) Find the coordinates of both stationary points. (b) Determine the nature of each stationary point using the second derivative $\frac{d^2y}{dx^2}$.

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16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Indices, Surds & Quadratics

The curve $y = x^2 - 6x + 5$ can be mapped to $y = x^2$ by a sequence of two geometric transformations. Express the equation in vertex form, state the vertex, and describe fully each transformation.



17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 10.01 Series & Binomial Theorem

A car of mass 1200 kg ascends a hill of inclination α, where $\sin\alpha = \frac{1}{20}$, against a constant resistance to motion of 500 N. Taking $g = 9.8$ m/s^2: (a) find the driving force required to travel at a steady speed of 25 m/s, (b) calculate the power developed by the engine in kilowatts.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 11.01 Kinematics

Events A and B are such that $P(A) = 0.6$, $P(B) = 0.5$, and $P(A \cup B) = 0.8$. Find $P(A \cap B)$, calculate $P(A|B)$, and determine whether A and B are independent.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 13.01 Newton's Laws

Find the equation of the perpendicular bisector of the line segment joining $A(-2, 5)$ and $B(4, -3)$. Give your answer in the form $ax + by + c = 0$, where a, b, c are integers.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 14.01 Energy, Work & Power

The random variable $X \sim N(\mu, \sigma^2)$ satisfies $P(X < 65) = 0.8413$ and $P(X < 45) = 0.0228$. Find the values of μ and σ.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A force of magnitude 50 N acts at an angle of 30° above the horizontal. Find its horizontal and vertical components.

a. Horizontal: 43.3 N, Vertical: 25.0 N

b. Horizontal: 25.0 N, Vertical: 43.3 N

c. Horizontal: 35.4 N, Vertical: 35.4 N

d. Horizontal: 50.0 N, Vertical: 0.0 N

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Two forces $F_1 = [8, -3]$ N and $F_2 = [-5, 7]$ N act on a particle. What is the magnitude of the resultant force?

a. 5 N

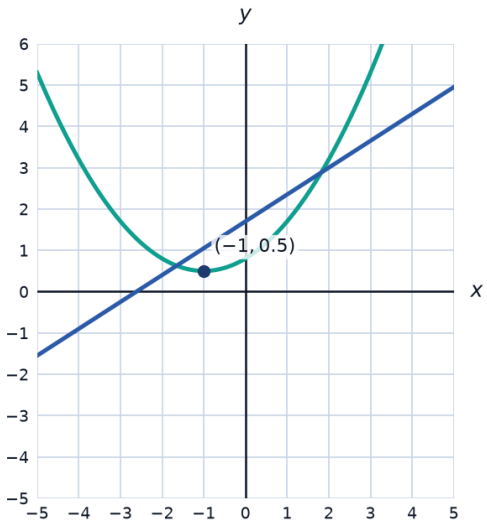
b. 7 N

c. $\sqrt{41}$ N

d. 12 N

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A particle is held in equilibrium by three coplanar forces: P acting East, Q acting North, and a force of 13 N acting at an angle θ South of West where $\tan\theta = 5/12$. Find the magnitude of P.



a. 12 N

b. 5 N

c. 13 N

d. 8 N

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A block of mass 6 kg rests on a rough horizontal table where the coefficient of friction is $\mu = 0.35$. Taking $g = 10$ m/s^2, what is the maximum frictional force that can act on the block?

a. 21 N

b. 60 N

c. 18 N

d. 24 N

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A horizontal force of 15 N is applied to the 6 kg block in the previous question (where $F_{\text{max}} = 21$ N). What is the actual frictional force acting on the block?

a. 15 N

b. 21 N

c. 0 N

d. 6 N

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A particle of mass 4 kg is placed on a rough plane inclined at 30° to the horizontal. If the particle is on the point of slipping down the plane, find the coefficient of friction μ.

a. $\mu = \tan(30^\circ) = \frac{\sqrt{3}}{3} \approx 0.577$

b. $\mu = \sin(30^\circ) = 0.5$

c. $\mu = \cos(30^\circ) \approx 0.866$

d. $\mu = 0.25$

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A crate of mass 10 kg is held in equilibrium on a smooth slope inclined at 30° to the horizontal by a force P acting parallel to the slope. Taking $g = 10$ m/s^2, find P.

a. 50 N

b. 86.6 N

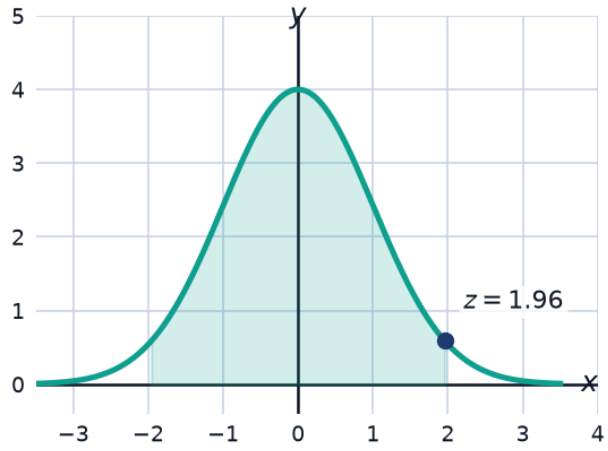
c. 100 N

d. 25 N

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the crate in the previous question, what is the normal contact force exerted by the slope on the crate?



a. 86.6 N ($50\sqrt{3}$ N)

b. 50 N

c. 100 N

d. 75 N

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A body of mass 8 kg rests in limiting equilibrium on an inclined plane. The normal reaction is 64 N. Taking $g = 10$ m/s^2, find the angle of inclination α of the plane.

a. $\alpha = \arccos(0.8) \approx 36.9^\circ$

b. $\alpha = 30^\circ$

c. $\alpha = 45^\circ$

d. $\alpha = 60^\circ$

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30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A lamp of weight $W = 60$ N is suspended by two light strings making angles of 30° and 60° with the horizontal ceiling. Find the tension in the steeper string.

a. $30\sqrt{3}$ N ≈ 52.0 N

b. 30 N

c. 60 N

d. 45 N

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A light inextensible string passes over a smooth peg. A mass of 5 kg hangs on one side and a mass of 3 kg hangs on the other. What downward external force must be applied to the 3 kg mass to maintain equilibrium? ($g = 10$ m/s^2)

a. 20 N

b. 50 N

c. 30 N

d. 10 N

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A ring of mass m is threaded on a rough horizontal wire with coefficient of friction $\mu = 0.4$. A light string attached to the ring is pulled with tension T at an angle of 45° above the wire. Find the value of T (in terms of mg) when the ring is in limiting equilibrium.

a. $T = \dfrac{0.4mg}{\cos(45^\circ) + 0.4\sin(45^\circ)} = \dfrac{0.4mg}{1.4\frac{\sqrt{2}}{2}} = \dfrac{4\sqrt{2}}{7}mg$

b. $\dfrac{\sqrt{2}}{2}mg$

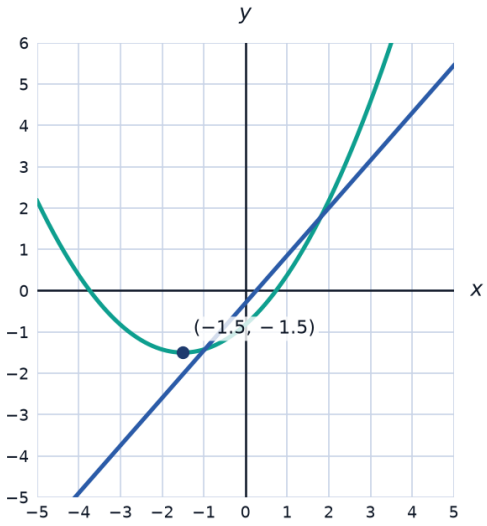
c. $0.4mg$

d. mg

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33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A particle of mass 5 kg rests on a smooth plane inclined at 45° to the horizontal. Taking $g = 9.8$ m/s^2, find the component of its weight acting down the slope, leaving your answer in exact (surd) form.



a. 49 N

b. $49/2 \times \frac{\sqrt{2}}{2}$ N

c. $51 \times \frac{\sqrt{2}}{2}$ N

d. $49 \times \frac{\sqrt{2}}{2}$ N

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A force of 10 N acts at 60° above the horizontal. Find its vertical component, leaving your answer in exact (surd) form.

a. $10 \times \frac{1}{2}$ N

b. $10 \times \frac{\sqrt{3}}{2}$ N

c. 10 N

d. $5 \times \frac{\sqrt{3}}{2}$ N

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A particle of mass 5 kg rests on a smooth plane inclined at 60° to the horizontal. Taking $g = 9.8$ m/s^2, find the normal reaction, leaving your answer in exact (surd) form.

a. $49 \times \frac{1}{2}$ N

b. $49/2 \times \frac{1}{2}$ N

c. $49 \times \frac{\sqrt{3}}{2}$ N

d. 49 N

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A force of 40 N acts at 60° above the horizontal. Find its horizontal component, leaving your answer in exact (surd) form.

a. 40 N

b. $20 \times \frac{1}{2}$ N

c. $40 \times \frac{\sqrt{3}}{2}$ N

d. $40 \times \frac{1}{2}$ N

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Two perpendicular forces of magnitude 6 N and 8 N act on a particle. Find the magnitude of the resultant force.

a. 14

b. 10

c. 8

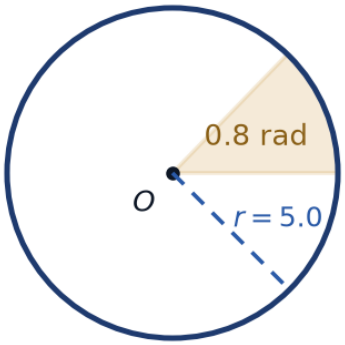
d. 12

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 11.01 Kinematics

A sector has radius 5 cm and angle 0.8 rad. Find the perimeter of the sector.



a. 14

b. 9

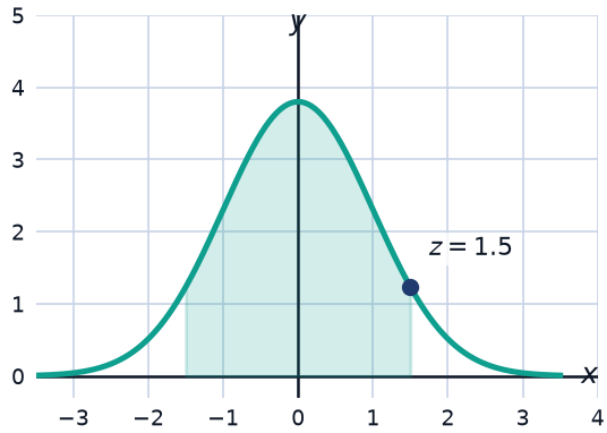
c. 4

d. 15

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 13.01 Newton's Laws

The scores on a standardized test are normally distributed with a mean of 100 and a standard deviation of 15. A new scoring algorithm is implemented, and the scores are transformed using the rule $Y = aX + b$, where X is the original score and Y is the new score. The new distribution of scores, Y, also follows a normal distribution with a mean of 120 and a standard deviation of 22.5. Calculate the value of a and the value of b.



a. a = 1.5, b = -30

b. a = -1.5, b = 270

c. a = 1.5, b = 30

d. a = 0.67, b = 53

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 14.01 Energy, Work & Power

The diagram shows the graph of the function $y = f(x)$. The function $f(x)$ is defined for all real numbers. Which of the following represents the graph of $y = f^{-1}(x)$?

a. Graph A

b. Graph B

c. Graph C

d. Graph D

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 15.01 Representation of Data

A research team is studying the impact of different fertilizer types on crop yield. They conducted an experiment with two fertilizer types, A and B, and recorded the yield in kilograms per plot. The data is summarized in the two-way table below. The total number of plots is 200. | Fertilizer Type | Yield < 50 kg | Yield $\geq$ 50 kg | Total | |---|---|---|---| | A | 30 | 70 | 100 | | B | 50 | 50 | 100 | | Total | 80 | 120 | 200 | Given that a plot was selected at random and its yield was less than 50 kg, what is the probability that it received fertilizer type B? Provide your answer as a fraction in its simplest form.

a. 1/2

b. 5/8

c. 3/8

d. 1/4

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 16.01 Permutations & Combinations

Describe fully the single transformation that maps $y = x^2$ to $y = (x - 4)^2$.

a. Reflection in the x-axis

b. Translation in the y-direction only

c. A stretch parallel to the y-axis

d. Translation by 4 unit(s) in the positive x-direction

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 17.01 Discrete Random Variables

A discrete random variable X has distribution $P(X = 1) = 2/5, P(X = 2) = 1/5, P(X = 3) = 2/5$. Given $Y = 5X - 3$, find $E(Y)$.

a. 6

b. 7

c. 8

d. 13

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44 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 18.01 Binomial Distribution**

Students are surveyed until the first one who owns a car is found; X is the number of students asked. Is X binomially distributed?

a. Yes — but only if $n > 30$

b. Yes — fixed, independent trials with constant success probability

c. No — the number of trials is not fixed

d. No — trials are not independent / probability is not constant

45 **[Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 19.01 Normal Distribution**

A machine's failure probability increases each hour it runs; X is the number of failures in a day. Is X binomially distributed?

a. No — trials are not independent / probability is not constant

b. Yes — fixed, independent trials with constant success probability

c. No — the number of trials is not fixed

d. Yes — but only if $n > 30$

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